Chapter 8 Intro to Networking Exam

Candidate: Dunkan Gibson (dunkan.gibson)
Date: 10/9/2021 4:56:08 pm • Time spent: 24:23

Score: 88% Passing Score: 80%

Question 1: ✔ Correct

You are configuring a firewall to allow access to a server hosted on the demilitarized zone of your network. You open TCP/IP ports 80, 25, 110, and 143.

Assuming that no other ports on the firewall need to be configured to provide access, which applications are most likely to be hosted on the server?

- Email server, Newsgroup server, and DNS server
- Web server, DNS server, and email server
- ➡ Web server and email server
- Web server, DNS server, and DHCP server

EXPLANATION

TCP/IP port 80 is associated with accessing web pages from a web server using the hypertext transfer protocol (HTTP). Email can be accessed using a number of protocols, including the simple mail transfer protocol (SMTP), the post office protocol version 3 (POP3) and the internet message access protocol version 4 (IMAP4). SMTP uses TCP/IP port 25, while POP3 uses TCP/IP port 110, and IMAP4 uses TCP/IP port 143.

Domain name service (DNS) traffic uses TCP/IP port 53. Newsgroup servers are accessed using the network news transfer (NNTP) protocol on TCP/IP port 119. Dynamic host configuration protocol (DHCP) traffic uses the BOOTP protocol on TCP/IP ports 67 and 68.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 2: ✔ Correct

Which of the following are characteristics of a packet filtering firewall? (Select two.)

- [ ] Filters by session
- [ ] Stateful
- [x] Filters IP addresses, not ports
- [ ] Filters by URL
- [x] Stateless

EXPLANATION

A packet filtering firewall makes decisions about which network traffic to allow by examining information in the IP packet header, such as source and destination addresses, ports, and service protocols. A packet filtering firewall is considered a stateless firewall because it examines each packet and uses rules to accept or reject each packet without considering whether the packet is part of a valid and active session.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 3: ✔ Correct

Which of the following features are common functions of an all-in-one security appliance? (Select two.)

- [ ] Quality of Service
- [x] Spam filtering
- [ ] Password complexity
- [ ] Content caching
- [x] Bandwidth shaping

EXPLANATION

All-in-one security appliances combine many security functions into a single device. Security functions in an all-in-one security appliance can include:

- Spam filter
- URL filter
- Web content filter
- Malware inspection
- Intrusion detection system

In addition to security functions, all-in-one security appliances can include:

- Network switch
- Router
- Firewall
- TX uplink (integrated CSU/DSU)
- Bandwidth shaping

REFERENCES

LabSim for Network Pro, Section 8.2.

Question 4: ✔ Correct

Which of the following is likely to be located in a DMZ?

- User workstations
- Backup server
- Domain controller
- FTP server

EXPLANATION

An FTP server is the most likely component from this list to be located in a DMZ (demilitarized zone) or a buffer subnet. A DMZ should only contain servers that are to be accessed by external visitors. Often it is assumed that any server placed in the DMZ will be compromised. Therefore, no mission critical or sensitive systems are located in a DMZ.

A domain controller may appear in a DMZ when the DMZ is an entire isolated domain, but this practice is not common. User workstations are never located in a DMZ. Unless specifically deployed for just the DMZ, backup servers are never located in a DMZ.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 5: ✔ Correct

You have a company network that is connected to the internet. You want all users to have internet access, but you need to protect your private network and users. You also need to make a web server publicly available to internet users.

Which solution should you use?

- ➡ Use firewalls to create a DMZ. Place the web server inside the DMZ and the private network behind the DMZ.
- Use a single firewall. Put the web server in front of the firewall and the private network behind the firewall.
- Use firewalls to create a DMZ. Place the web server and the private network inside the DMZ.
- Use a single firewall. Put the web server and the private network behind the firewall.

EXPLANATION

A demilitarized zone (DMZ), also called a screened subnet, is a buffer network (or subnet) that sits between the private network and an untrusted network, such as the internet. A common configuration uses two firewalls, one connected to the public network, and one connected to the private network. Publicly accessible resources (servers) are placed inside the screened subnet. Examples of publicly accessible resources include web, FTP, or email servers. Private resources that are not accessible from the internet are placed behind the DMZ (behind the inner firewall).

Placing the web server inside the private network would mean opening ports in the firewall leading to the private network, which could expose other devices to attack. Placing the web server outside of the firewall would leave it unprotected.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 6: ✔ Correct

You manage a small network at work. Users use workstations connected to your network. No portable computers are allowed.

As part of your security plan, you would like to scan all users' emails. You want to scan the emails and prevent any e-mails with malicious attachments from being received by users.

Your solution should minimize administration, allowing you to centrally manage the scan settings.

Which solution should you use?

- SMTP
- DMZ
- Host-based firewall
- ➡ Network-based firewall

EXPLANATION

A network-based firewall inspects traffic as it flows between networks. For example, you can install a network-based firewall on the edge of your private network that connects to the internet and scans all incoming emails. Scanning emails as they arrive at your email server allows you to centralize management and stop malicious emails before they arrive at client computers.

A demilitarized zone (DMZ), also called a screened subnet, is a buffer network, or subnet, that sits between the private network and an untrusted network, such as the internet. SMTP is an email protocol used by email servers for sending mail.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 7: ✕ Incorrect

Match the firewall type on the right with the OSI layer at which it operates.

Each OSI Layer may be used once, more than once, or not at all.

Packet filtering firewall	Circuit-level proxy	Application-level gateway
✓ OSI Layer 3	✓ OSI Layer 5	✓ OSI Layer 7

Routed firewall	Transparent firewall
~~OSI Layer 2~~ OSI Layer 3	~~OSI Layer 3~~ OSI Layer 2

EXPLANATION

Each firewall type operate at a specific layer of the OSI model.

- Packet filtering firewalls operate at Layer 3.
- Circuit-level proxies operate at Layer 5.
- Application-level gateways operate at Layer 7.
- Routed firewalls operate at Layer 3.
- Transparent firewalls operate at Layer 2.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 8: ✔ Correct

You want to install a firewall that can reject packets that are not part of an active session.

Which type of firewall should you use?

- Packet filtering
- Application-level
- VPN concentrator
- ➡ Circuit-level

EXPLANATION

A circuit-level proxy, or circuit-level gateway, makes decisions about which traffic to allow based on virtual circuits or sessions. A circuit-level gateway:

- Operates at OSI Layer 5 (Session layer).
- Keeps a table of known connections and sessions. Packets directed to known sessions are accepted.
- Verifies that packets are properly sequenced.
- Ensures that the TCP three-way handshake process occurs only when appropriate.
- Does not filter packets; rather, it allows or denies sessions.

A packet filtering firewall makes decisions about which network traffic to allow by examining information in the IP packet header, such as source and destination addresses, ports, and service protocols. An Application level gateway is a firewall that is capable of filtering by information contained within the data portion of a packet, such as URLs within an HTTP request. A VPN concentrator is a device that is used to establish remote access VPN connections.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 9: ✔ Correct

Which of the following does a router acting as a firewall use to control which packets are forwarded or dropped?

- ➡ ACL
- IPsec
- PPP
- RDP
- VNC

EXPLANATION

When you configure a router as a firewall, you configure the access control list (ACL) with statements that identify traffic characteristics, such as the direction of traffic (inbound or outbound), the source or destination IP address, and the port number. ACL statements include an action to either allow or deny the traffic specified by the ACL statement.

IPsec is a protocol for encrypting packets. RDP and VNC are remote desktop protocols used for remotely accessing a computer's desktop. PPP is a protocol for establishing a remote access connection over a dial-up link.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 10: ✔ Correct

Match the firewall type on the left with its associated characteristics on the right. Each firewall type may be used once, more than once, or not at all.

Operates at Layer 2.

✔ Virtual firewall

Operates at Layer 3.

✔ Routed firewall

Counts as a hop in the path between hosts.

✔ Routed firewall

Does not count as a hop in the path between hosts.

✔ Virtual firewall

Each interface connects to a different network.

✔ Routed firewall

Each interface connects to the same network segment.

✔ Virtual firewall

EXPLANATION

In a routed firewall, the firewall is also a Layer 3 router. In fact, many hardware routers include firewall functionality. Transmitting data through this type of firewall counts as a router hop. A routed firewall usually supports multiple interfaces, each connected to a different network segment.

A transparent firewall (also called a virtual firewall) works differently. It operates at Layer 2, and it is not seen as a router hop by connected devices. Both the internal and external interfaces on a transparent firewall connect to the same network segment. Because it is not a router, you can easily introduce a transparent firewall into an existing network.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 11: ✔ Correct

You manage a server that runs your company website. The web server has reached its capacity, and the number of client requests is greater than the server can handle.

You would like to find a solution so that static web content can be offloaded to a different server while the web server continues to process dynamic content.

Which solution should you implement?

- Ethernet bonding
- ➡ Proxy server
- Traffic shaper
- QoS

EXPLANATION

A common application of a caching engine on a network is a proxy server configured to cache web content. As users visit websites, web content is retrieved from the web server and cached on the proxy server. Subsequent requests for the same content are handled by the proxy server from cache rather than retrieved from the web server.

A traffic shaper (also called a bandwidth shaper) is a device that is capable of modifying the flow of data through a network in response to network traffic conditions. Quality of Service (QoS) refers to a set of mechanisms that try to guarantee timely delivery or minimal delay of important or time-sensitive communications. QoS is particularly important if you implement Voice over IP (VoIP), Video over IP, or online games where delay or data loss make the overall experience unacceptable.

Ethernet bonding (also called NIC teaming), logically groups two or more physical connections to the same network. Data is divided and sent on multiple interfaces, effectively increasing the speed at which the device can send and receive on the network.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 12: ✔ Correct

You have been given a laptop to use for work. You connect the laptop to your company network, use it from home, and use it while traveling.

You want to protect the laptop from internet-based attacks.

Which solution should you use?

- ➡ Host-based firewall
- Proxy server
- VPN concentrator
- Network-based firewall

EXPLANATION

A host-based firewall inspects traffic received by a host. Use a host-based firewall to protect your computer from attacks when there is no network-based firewall, such as when you connect to the internet from a public location.

A network-based firewall inspects traffic as it flows between networks. For example, you can install a network-based firewall on the edge of your private network that connects to the internet to protect your data from attacks from internet hosts.

A VPN concentrator is a device connected to the edge of a private network that is used for remote access VPN connections. Remote clients establish a VPN connection to the VPN concentrator and are granted access to the private network. A proxy server is an Application layer firewall that acts as an intermediary between a secure private network and the public. Access to the public network from the private network goes through the proxy server.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 13: ✔ Correct

Which firewall implementation creates a buffer network that can be used to host email or web servers?

- Host-based firewall
- Perimeter firewall
- ➡ DMZ
- ACL

EXPLANATION

A demilitarized zone (DMZ), also called a screened subnet, is a buffer network (or subnet) that sits between the private network and an untrusted network (such as the internet). Web servers and email servers can be placed in this buffer network so that they can be accessed from the internet.

Firewalls use access control lists (ACLs) to manage incoming or outgoing traffic. A perimeter firewall sits on the edge of a private network and filters all incoming and outgoing requests. A host-based firewall is used to protect a single device.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 14: ✔ Correct

Which of the following network services or protocols uses TCP/IP port 22?

- TFTP
- NNTP
- IMAP4
- SSH

EXPLANATION

The Secure Shell (SSH) service uses TCP/IP port 22. SSH is a terminal emulation program similar to Telnet, which provides secure authenticated sessions on a remote system. It is most commonly associated with Unix and Linux systems.

The trivial file transfer protocol (TFTP) is a connectionless service for downloading files from a remote system. TFTP uses TCP/IP port 69. The network news transfer protocol (NNTP) is used to access and download messages from newsgroup servers. NNTP uses TCP/IP port 119. The internet message access protocol version 4 (IMAP4) is used to download email from remote servers. IMAP 4 uses TCP/IP port 143.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 15: ✕ Incorrect

To answer this question, complete the lab using the information below. **Launch Lab**

You did not attempt the lab.

Question 16: ✔ Correct

After blocking a number of ports to secure your server, you are unable to send email. To allow email service, which of the following needs to be done?

- Open port 80 to allow SNMP service.
- Open port 110 to allow SMTP service.
- Open port 110 to allow POP3 service.
- Open port 80 to allow SMTP service.
- Open port 25 to allow SNMP service.
- ➡ Open port 25 to allow SMTP service.

EXPLANATION

The simple mail transfer protocol (SMTP) uses TCP port 25 and is responsible for sending email. If port 25 is blocked, users will not be able to send email, but they could receive email using port 110 and the POP3 protocol.

SNMP is used to monitor network traffic. POP3 uses port 110 and is used to retrieve email from a mail server.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 17: ✔ Correct

Which of the following are true of a circuit proxy filter firewall? (Select two.)

- [ ] Operates at ring 0 of the operating system.
- [ ] Operates at the Network and Transport layers.
- [ ] Examines the entire message contents.
- [x] Operates at the Session layer.
- [x] Verifies sequencing of session packets.
- [ ] Operates at the Application layer.

EXPLANATION

A circuit proxy filter firewall operates at the Session layer. It verifies the sequencing of session packets, breaks the connections, and acts as a proxy between the server and the client.

An Application layer firewall operates at the Application layer, examines the entire message, and can act as a proxy to clients. A stateful inspection firewall operates at the Network and Transport layers. It filters on both IP addresses and port numbers. A kernel proxy filtering firewall operates at the operating system ring 0.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 18: ✔ Correct

Which of the following are true about reverse proxy? (Select two.)

- [x] Can perform load balancing, authentication, and caching.
- [ ] Clients always know they are using reverse proxy.
- [x] Handles requests from the internet to a server in a private network.
- [ ] Handles requests from inside a private network out to the internet.
- [ ] Sits between a client computer and the internet.

EXPLANATION

A reverse proxy server handles requests from the internet to a server located inside a private network. Reverse proxies can perform load balancing, authentication, and caching.

Reverse proxies often work transparently, meaning clients don't know they are connected to a reverse proxy.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 19: ✔ Correct

Which of the following is the best device to deploy to protect your private network from a public untrusted network?

- Gateway
- Hub
- Router
- ➡ Firewall

EXPLANATION

A firewall is the best device to deploy to protect your private network from a public untrusted network. Firewalls are used to control traffic entering and leaving your trusted network environment. Firewalls can manage traffic by source or destination IP address, port number, service protocol, application or service type, user account, and even traffic content.

Routers offer some packet-based access control, but not as extensive as that of a full-fledged firewall. Hubs and gateways are not sufficient for managing the interface between a trusted network and an untrusted network.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 20: ✕ Incorrect

Which protocol and port number is used by TFTP?

- UDP 69
- TCP 20 and 21
- TCP 67
- UDP 67
- ~~TCP 69~~
- UDP 20 and 21

EXPLANATION

The trivial file transfer protocol (TFTP) uses UDP port 69. TFTP is a faster alternative to FTP. TFTP uses UDP instead of TCP to reduce the overhead of sending packets.

FTP uses TCP ports 20 and 21. DHCP uses port 67.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 21: ✔ Correct

Haley configures a website using Windows Server 2016 default values.

What are the HTTP port and SSL port settings?

- ➡ 80 for HTTP; 443 for SSL
- 160 for HTTP; 440 for SSL
- 443 for HTTP; 80 for SSL
- 440 for HTTP; 160 for SSL

EXPLANATION

The default TCP port setting for HTTP is 80. You can change that setting to another TCP setting that is not in use, but users will have to know they must request the non-default setting, or they will be unable to connect. The SSL port number is 443 and is only used with secure socket layers for encryption.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 22: ✔ Correct

A small startup company has hired you to harden their new network.

Because funds are limited, you have decided to implement a unified threat management (UTM) device that provides multiple security features in a single network appliance:

- Firewall
- VPN
- Anti-spam
- Antivirus

You join the UTM device to the company's Active Directory domain. The company's traveling sales force will use the VPN functionality provided by the UTM device to connect to the internal company network from hotel and airport public Wi-Fi networks.

Which weaknesses exist in this implementation?

- The UTM device should not be joined to the Active Directory domain.
- ➡ The UTM represents a single point of failure.
- Anti-spam and antivirus functions must be placed on separate servers.
- Remote users should not establish VPN connections to the internal network using public Wi-Fi networks.

EXPLANATION

A unified threat management device combines multiple security features into a single network appliance. A single UTM device can provide several security features:

- Firewall
- VPN
- Ant-spam
- Anti-virus
- Load balancing

By combining several services into one appliance, UTM devices make managing network security less expensive and complex. However, they also introduce a single point of failure—if the UTM fails, then network security is lost. Additionally, UTM devices aren't as robust as other devices made for a specific use. Therefore, UTM devices are best suited for:

- Offices where space limits don't allow multiple security appliances.
- Satellite offices that need to be managed remotely. Configuration changes need to be made on only one device, rather than multiple devices.
- Smaller businesses that wouldn't benefit from the robust features provided by specific security appliances.

Joining the UTM device to the Active Directory domain is the best practice because it simplifies authentication management. Using a VPN connection while accessing a public Wi-Fi network is a good practice because it provides an extra layer of security. Since this organization is a small startup, combining anti-spam and anti-virus functions on the same device is an acceptable practice.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 23: ✔ Correct

Which protocols and port numbers are used by DHCP? (Select two.)

- [ ] UDP 123
- [x] UDP 67
- [x] TCP 67
- [ ] TCP 123
- [ ] UDP 69
- [ ] TCP 69

EXPLANATION

The dynamic host configuration protocol (DHCP) uses port 67 for both the TCP and UDP protocols. TFTP uses UDP port 69, and NTP uses UDP port 123.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 24: ✔ Correct

You connect your computer to a wireless network available at the local library. You find that you can access all the websites you want on the internet except for two.

What might be causing the problem?

- Port triggering is redirecting traffic to the wrong IP address.
- A firewall is blocking ports 80 and 443.
- ➡ A proxy server is blocking access to the websites.
- The router has not been configured to perform port forwarding.

EXPLANATION

A proxy server can be configured to block internet access based on website or URL. Many schools and public networks use proxy servers to prevent access to websites with objectionable content.

Ports 80 and 443 are used by HTTP to retrieve all web content. If a firewall were blocking these ports, access would be denied to all websites. Port forwarding directs incoming connections to a host on the private network. Port triggering dynamically opens firewall ports based on applications that initiate contact from the private network.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 25: ✔ Correct

Which of the following are true about routed firewalls? (Select two.)

- [x] Supports multiple interfaces.
- [ ] Operates at Layer 2.
- [ ] Easily introduced to an existing network.
- [ ] Internal and external interfaces connect to the same network segment.
- [x] Counts as a router hop.

EXPLANATION

In a routed firewall, the firewall is also a Layer 3 router. In fact, many hardware routers include firewall functionality. Transmitting data through this type of firewall counts as a router hop. A routed firewall usually supports multiple interfaces, each connected to a different network segment.

A transparent firewall (which is also called a virtual firewall) works differently. It operates at Layer 2 and is not seen as a router hop by connected devices. Both the internal and external interfaces on a transparent firewall connect to the same network segment. Because it is not a router, you can easily introduce a transparent firewall into an existing network.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 26: ✔ Correct

You want to allow users to download files from a server running the TCP/IP protocol. You want to require user authentication to gain access to specific directories on the server.

Which TCP/IP protocol should you implement to provide this capability?

- IP
- TFTP
- HTTP
- HTML
- ➡ FTP
- TCP

EXPLANATION

You should implement the file transfer protocol (FTP). It enables file transfers and supports user authentication. The trivial file transfer protocol (TFTP) enables file transfer, but does not support user authentication.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 27: ✔ Correct

You want to transfer a file from a UNIX server to Windows Server 2016.

Which of the following utilities could you use to do this? (Select all that apply.)

- ➡ ☑ TFTP
- ☐ Tracert
- ➡ ☑ FTP
- ☐ Netstat
- ☐ NBTSTAT
- ☐ Telnet

EXPLANATION

UNIX computers and Windows Server 2016 systems use TCP/IP. Therefore, the TCP/IP utilities FTP and TFTP both allow you to transfer files.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 28: ✔ Correct

Which of the following is a firewall function?

- ○ Frame filtering
- ➡ ◉ Packet filtering
- ○ Protocol conversion
- ○ Encrypting
- ○ FTP hosting

EXPLANATION

Firewalls often filter packets by checking each packet against a set of administrator-defined criteria. If the packet is not accepted, it is simply dropped.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 29: ✔ Correct

Which of the following functions are performed by proxies? (Select two.)

- [ ] Block unwanted packets from entering your private network
- [x] Block employees from accessing certain websites
- [ ] Give users the ability to participate in real-time text-based internet discussions
- [ ] Filter unwanted email
- [ ] Store client files
- [x] Cache web pages

EXPLANATION

A proxy, or proxy server, stands between client computers and internet web servers. You can use a proxy server to prevent access to specific websites or to cache (save) frequently used web pages. When a proxy receives a request from the client, it checks to verify that the client is allowed access to the website. If allowed, it then checks its cache to see if the requested page is in the cache. If the page is already cached, then the proxy server fulfills the request by displaying the requested page from the cache rather than retrieving it from the internet. Receiving a web page from a local proxy server is much faster than downloading the page from the internet.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 30: ✔ Correct

Which of the following is a firewall function?

- Encrypting
- ➡ Packet filtering
- Protocol conversion
- FTP hosting
- Frame filtering

EXPLANATION

Firewalls often filter packets by checking each packet against a set of administrator-defined criteria. If the packet is not accepted, it is simply dropped.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 31: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 32: ✔ Correct

You want to maintain tight security on your internal network, so you restrict access to the network through certain port numbers. If you want to allow users to continue to use DNS, which port should you enable?

- 42
- 443
- 80
- 53
- 21

EXPLANATION

The DNS service uses port 53.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 33: ✔ Correct

You are monitoring network traffic on your network, and you see traffic between two network hosts on port 2427.

Which kind of network traffic uses this port?

- A ping of death attack on a network host is in progress.
- A workstation is using the DHCP protocol to request an IP address from a DHCP server.
- ➡ The MGCP protocol is generating traffic, which VoIP uses to send voice data over a network.
- Someone is remotely accessing another system using the SSH protocol.

EXPLANATION

Someone on the network is using voice over IP (VoIP) to make a telephone call. Some VoIP implementations use the media gateway control protocol (MGCP) to set up, maintain, tear down, and redirect calls. MGCP uses port 2427.

The DHCP protocol is used to automatically assign IP addresses to network hosts and utilizes IP ports 67 and 68. A ping of death attack utilizes an oversized ICMP echo request packet to crash a target system. The SSH protocol is used to remotely access another network host and uses port 22.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 34: ✔ Correct

Which of the following solutions would you implement to track which websites network users are accessing?

- ➡ (●) Proxy
- ( ) Packet filtering firewall
- ( ) NIDS
- ( ) Tarpit

EXPLANATION

A proxy server is a device that stands as an intermediary between a secure private network and the public. It is a specific implementation of an Application layer firewall. A proxy stops and inspects every packet at the firewall, which causes a break between the client and the source server. Proxies can be configured to filter content, monitor internet access, or restrict access by user or by website.

A packet filtering firewall examines the packet header information to make forwarding decisions. The firewall can accept or reject packets based on IP address, but not individual websites.

A network-based IDS (NIDS) is a dedicated device installed on the network. It analyzes all traffic on the network to look for potential attacks. A tarpit (also called a sticky honeypot) is a honeypot that answers connection requests in such a way that the attacking computer is stuck for a period of time.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 35: ✔ Correct

Which of the following prevents access based on website ratings and classifications?

- ➡ Content filter
- NIDS
- DMZ
- Packet-filtering firewall

EXPLANATION

An internet content filter is software used to monitor and restrict what content is delivered to an end user across the web. Companies, schools, libraries, and families commonly use content filters to restrict internet access, block specific websites, or block specific content.

A packet-filtering firewall examines the packet header information to make forwarding decisions. The firewall can accept or reject packets based on IP address, but not individual websites. A network-based IDS (NIDS) is a dedicated device installed on the network. It analyzes all traffic on the network to look for potential attacks. A demilitarized zone (DMZ) is a buffer network (or subnet) that sits between the private network and an untrusted network (such as the internet).

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 36: ✔ Correct

You have a router that is configured as a firewall. The router is a Layer 3 device only.

Which of the following does the router use for identifying allowed or denied packets?

- Session ID
- Username and password
- ➡ IP address
- MAC address

EXPLANATION

A router acting as a firewall at Layer 3 is capable of making forwarding decisions based on the IP address.

The MAC address is associated with OSI model layer 2. Switches and wireless access points use MAC addresses to control access. The session ID is used by a circuit-level gateway, and username and password are used by Application layer firewalls.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 37: ✔ Correct

You are configuring a network firewall to allow SMTP outbound email traffic and POP3 inbound email traffic.

Which of the following TCP/IP ports should you open on the firewall? (Select two.)

- [ ] 443
- [x] 110
- [ ] 21
- [ ] 143
- [x] 25

EXPLANATION

The simple mail transfer protocol (SMTP) uses TCP/IP port 25. The post office protocol version 3 (POP3) uses TCP/IP port 110.

The file transfer protocol (FTP) uses TCP/IP Port 21. The internet message access protocol (IMAP) uses TCP/IP port 143. TCP/IP port 443 is used by the secure sockets layer (SSL) protocol.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 38: ✔ Correct

Which of the following describes how access lists can be used to improve network security?

- ➡ An access list filters traffic based on the IP header information such as source or destination IP address, protocol, or socket numbers.
- An access list identifies traffic that must use authentication or encryption.
- An access list filters traffic based on the frame header such as source or destination MAC address.
- An access list looks for patterns of traffic between multiple packets and takes action to stop detected attacks.

EXPLANATION

An access list filters traffic based on the IP header information such as source or destination IP address, protocol, or socket numbers. Access lists are configured on routers and operate on Layer 3 information.

Port security is configured on switches and filters traffic based on the MAC address in the frame. An intrusion detection system (IDS) or intrusion prevention system (IPS) examines patterns detected across multiple packets. An IPS can take defensive action when a suspicious pattern of traffic is detected.

REFERENCES

LabSim for Network Pro, Section 8.3.

Question 39: ✕ Incorrect

Which of the following is a file transfer protocol that uses UDP?

- ~~FTP~~
- POP3
- SMTP
- HTTP
- TFTP
- NTP

EXPLANATION

The trivial file transfer protocol (TFTP) uses UDP. The file transfer protocol (FTP) is a file transfer protocol, but uses TCP instead of UDP.

HTTP is the hypertext transfer file. It is used for web browsing and is not considered a file transfer protocol. SMTP and POP3 are email transfer protocols. NTP is a service that provides network time.

REFERENCES

LabSim for Network Pro, Section 8.1.

Question 40: ✕ Incorrect

To answer this question, complete the lab using the information below. **Launch Lab**

You did not attempt the lab.

Copyright © 2021 TestOut Corporation All rights reserved.